

Sheet 3 - Knowledge

outstanding

1a) - we have two ISNE of (T, L) and (B, R), by underlining BR payoffs.

- At MSNE, let p^1, p^2 play T, L w.p. p, q . Then by indifference principle

$$\Rightarrow q = \frac{3(1-q)}{3/4} \text{ and } 3p = 1-p$$

so we have $(\frac{1}{4}T + \frac{3}{4}B, \frac{3}{4}L + \frac{1}{4}R)$
and π payoffs $(\frac{3}{4}, \frac{3}{4})$

b) A BNE is a NE of the derived strategic game where we add artificial players for each (player, type) pair in the original game. [Hmm, though here we have continuous types...]

We're looking for $\varepsilon_1^*, \varepsilon_2^*$ s.t. p^1 plays B iff $\varepsilon_1 \geq \varepsilon_1^*$; p^2 plays L iff $\varepsilon_2 \geq \varepsilon_2^*$. For these strategies to be BNE they must be BRs to the opponent's strategy. ~~Retracts~~
to this strat. of p^2

For p^1 , B is a BR whenever $v_1(B) \geq v_1(T)$, i.e.

$$(1 - \varepsilon_2^*/\bar{\varepsilon}) \leq (3 + \varepsilon_1) \cdot \varepsilon_2^*/\bar{\varepsilon}$$

$$\Rightarrow \varepsilon_1 \geq \bar{\varepsilon}/\varepsilon_2^* - 4 =: \varepsilon_1^*$$

Good

And similarly for p^2 , T is a BR to p^1 's strat whenever $\varepsilon_2 \geq \bar{\varepsilon}/\varepsilon_1^* - 4 =: \varepsilon_2^*$

Does this carry across to BNE?

Since this is a symmetric game, it has a symmetric eq*. So we can impose that $\varepsilon_1^* = \varepsilon_2^*$ and solve.

↳ You can conclude that $\varepsilon_1^* = \varepsilon_2^*$ from the two conditions above.
 $\bar{\varepsilon} - 4\varepsilon_1^* = \varepsilon_1^*\varepsilon_2^* = \bar{\varepsilon} - 4\varepsilon_2^* \Rightarrow \varepsilon_1^* = \varepsilon_2^*$

I guess we're claiming the derived game will be symmetric and so have a symm. NE?

So, $\bar{\epsilon}/\epsilon^* - 4 = \epsilon^*$

$$\epsilon^{*2} + 4\epsilon^* - \bar{\epsilon} = 0 \Rightarrow \epsilon^* = \frac{-4 \pm \sqrt{16 + 4\bar{\epsilon}}}{2}$$

$$= -2 \pm \sqrt{4 + \bar{\epsilon}}$$

but $\epsilon^* > 0$ so

$$\epsilon_1^* = \epsilon_2^* = \epsilon^* = -2 + \sqrt{4 + \bar{\epsilon}} \text{ forms the BNE in pure strats described.}$$

[interpretation of this?]

c) As $\bar{\epsilon} \rightarrow 0$, we have $\epsilon^* \rightarrow 0$, and

this alone, not much. It's how fast it goes to zero relative to $\bar{\epsilon}$ that matters.

~~$$Pr(B) = 2 \frac{\epsilon^*}{\bar{\epsilon}} = 2 \frac{-2 + \sqrt{4 + \bar{\epsilon}}}{\bar{\epsilon}}$$~~

~~$$Pr(B) = 2 \frac{\epsilon^*}{\bar{\epsilon}} = 2 \frac{-2 + \sqrt{4 + \bar{\epsilon}}}{\bar{\epsilon}}$$~~

$$Pr(B) = 1 - \frac{\epsilon_1^*}{\bar{\epsilon}} = 1 - \frac{-2 + \sqrt{4 + \bar{\epsilon}}}{\bar{\epsilon}}$$

$$\rightarrow 1 - \frac{\frac{1}{2} \cdot (4 + 0)^{-1/2}}{1} = 3/4 \text{ as } \bar{\epsilon} \rightarrow 0.$$

by L'Hôpital

and similarly $Pr(L) \rightarrow 3/4$.

So, the induced probability distributions over the pure strategies in the perturbed eq^s converge to the mixing probabilities in the unperturbed game of (a).

d) There's some justification provided - in real-life games, there's never truly complete information. So, it's reasonable to think of each player as having some stochastic private info and conditionally playing ~~deterministic~~ pure strategies but which to an outside observer looks like mixing. This resolves the objection that it's hard to explain why players would mix randomly in just the right way given indifference between included actions.

Nice

But it's unclear why we should think tiny, independent perturbations are the most realistic or predictive. Presumably, a different perturbation structure might converge to something else entirely.

2a) We can model the striker as receiving a perfect signal of their type $t \in \{sr, sl\}$ and the goalie getting the same signal in all states.

Then we have BR functions as follows:

	BR_{sr}			BR_{sl}	
L	R	$1 > 0.6$	L	R	$0.8 > 0.7$
r	L	$0.8 > 0.7$	r	L	$1 > 0.6$

For goalkeeper, BR maximises E payoff given their 50:50 beliefs:

$sr \searrow sl \swarrow$	$2v_g(L)$	$2v_g(r)$	BR_g
(L, L)	$0.3 + 0.4 = 0.7$	$0 + 0.2 = 0.2$	L
(L, R)	$0.4 + 0.2 = 0.6$	$0.2 + 0.4 = 0.6$	indiff
(R, L)	$0 + 0.3 = 0.3$	$0.3 + 0 = 0.3$	indiff.
(R, R)			r (as row 1)

So if in our PSBNE, G played L then S would play (R, R) . But L is not a BR to that. Similarly, $BR_g(BR_s(r)) = BR_g(L, L) = L$. So there's no PSBNE, since each player in the derived game must play a BR to the other's actually-played strategy (for NE).

b) Let G play L w.p. p . We can search for eq^* by indiff. principle and checking BR consistency.

• (L, L) and (R, R) cannot be part of this MSBNE since G is not indifferent between their actions in response, so wouldn't mix.

• For S to play (L, R) we require

$$BR_{sr} \quad 0.6p + 0.8(1-p) \geq p + 0.7(1-p) \quad \text{and}$$

$$BR_{sl} \quad 0.8p + 0.6(1-p) \geq 0.7p + 1-p$$

$$\Rightarrow 0.5p \leq 0.1$$

$$\text{and } 0.5p \geq 0.4 \Rightarrow \text{No } p \text{ solves this.}$$

o For (R, L),

$$BR_{SR} \quad p + 0.7(1-p) \geq 0.6p + 0.8(1-p)$$

$$BR_{SC} \quad 0.7p + 1-p \geq 0.84p + 0.6(1-p)$$

$$\Rightarrow 0.5p \geq 0.1 \quad \text{and} \quad 0.5p \leq 0.4$$

$$\text{So any } p \in [0.2, 0.8]$$

The relevant eq^s are $((R, L), pL + (1-p)R)$ for $p \in [0.2, 0.8]$

- c) G would have to mix, for ^{either} S to mix, since as shown in (a) $S_{L,R}$ has strict BRs against pure strats from G. ☹

So let G play L w.p. p .

If both types mix, they must both be indifferent between L and R given p . But there's no single p which satisfies this, since ~~one type would need~~ (like in (b)) we'd need $p = 0.2$ and $p = 0.8$. ✗

Now suppose only ^{SC} mixes; ~~then we'd play & since that's~~ we have $p = 0.8 > 0.2$. So a BR for SC is to play R. We're at some combination between (R, L) and (R, R) - but G is indifferent only at (R, L) (BRs linear so only cross once). So G plays a pure strategy - but then it's not BR for S to mix! ✗, and same argument if only SR mixes.

∴ No MSBNE where S mixes, in either type.

3. a) All ^{profiles} pure strategies $(d, 1-d)$ for $d \in [0, 1]$ are PSNE. Neither player has a strict unilateral incentive to deviate.

$$(+ \quad x=y=1)$$

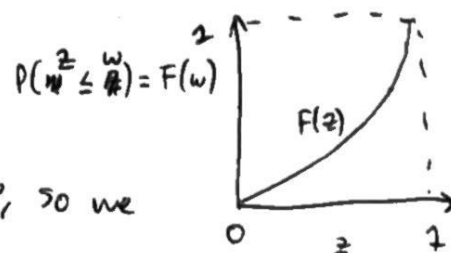
weird, but true.

b)
$$v_x = x^\alpha \cdot P\left(\frac{z \geq x+y}{x+y} \mid z \leq x+y\right) + 0 \cdot P\left(\frac{z < x+y}{x+y} \mid z \leq x+y\right).$$

 where $F(z) = z^n$ for $0 \leq z \leq 1$ and $n > 0$.

$$= x^\alpha \cdot (1 - (x+y)^n)$$

And similarly $v_y = y^\alpha \cdot (1 - (x+y)^n)$



c) Each player chooses their own strategic variable, so we require $\partial v_x / \partial x = 0$ and $\partial v_y / \partial y = 0$.

$$\Rightarrow \left. \begin{aligned} \alpha x^{\alpha-1} (1 - (x+y)^n) - n x^\alpha (x+y)^{n-1} &= 0 \\ \text{and } \alpha y^{\alpha-1} (1 - (x+y)^n) - n y^\alpha (x+y)^{n-1} &= 0 \end{aligned} \right\} \text{ at optimum.}$$

It wouldn't be optimum to demand zero since MU from demanding ϵ more is infinite, if paid, and $DP(\text{paid})$ is finite.

So divide both sides by $\{x, y\}^{\alpha-1}$.

$$\alpha (1 - (x+y)^n) = n \cdot x (x+y)^{n-1} = n \cdot y (x+y)^{n-1} \quad \text{(i) (ii)}$$

From (ii), $x^* = y^*$, so $\alpha (1 - (2x^*)^n) = n \frac{x^* (2x^*)^{n-1}}{= \frac{1}{2} (2x^*)^n}$

$$\alpha = \frac{1}{2} (2x^*)^n \cdot \left(\alpha + \frac{1}{2} n \right)$$

$$x^* = y^* = \frac{1}{2} \left(\frac{2\alpha}{2\alpha + n} \right)^{1/n} \quad \text{as required.}$$

d) A uniform distribution means $F(z) = z$, i.e. $n=1$.

So the optimal demands are

$$x^* = y^* = \frac{\alpha}{2\alpha + 1}, \quad \text{where } \alpha \text{ controls risk-aversion.}$$

For $\alpha=1$, they're risk-neutral. As $\alpha \rightarrow 0$, risk aversion increases $\rightarrow$ and demand $\downarrow$
 (these are CRRA preferences), and in the limit they demand 0. (i.e. demand decreases in risk-aversion as expected, since they prefer to avoid $P(\text{no payment})$ than increase payout in worlds where they get something.)

e) $\log(x^*) = \log(1/2) + \frac{1}{n} (\log(2\alpha) - \log(2\alpha + n))$; as $n \rightarrow \infty$

let $f(n) := \ln 2\alpha - \ln(2\alpha + n)$; $g(n) := n$.

$f'(n) = -\frac{1}{2\alpha + n}$, $g'(n) = 1$. So by L'Hôpital, $\log(x^*) \rightarrow \ln \frac{1}{2} + 0$ as $n \rightarrow \infty$. So $x^* \rightarrow \frac{1}{2}$.

As the probability distr of z becomes concentrated around 1, players' bids approach $\frac{1}{2}$, splitting the ~~expected~~ almost-sure pot equally between them. This makes sense: bidding any less would not increase $P(\text{get money})$ but reduce payout; bidding more means $P(\text{paid}) = 0$ from symmetry.

[Human, but maybe still profiles like (0.7, 0.3) would be NE, like in (a) !]

↳ ah, nevermind, no atoms so this has $P(\text{paid}) = 0$. ~~And thus~~ The FOCs pin down unique BRs and we only have one NE, not a continuum.

4. a) $\Omega = \{K, B, Ch\} \times \{cb, ncb\}$
for source $\times$ Charlie's colour-blindness.

b) $P_A = \{ \{Ch-cb, Ch-ncb, K-cb, K-ncb\}, \{B-cb, B-ncb\} \}$

$P_C = \{ \{Ch-cb, K-cb\}, \{B-cb\}, \{Ch-ncb\}, \{K-ncb\}, \{B-ncb\} \}$

c) ~~Let~~ Let $E = \{B-cb, B-ncb\}$.

Then $K_C(E) = \{B-cb, B-ncb\}$ since in both cases $P_C(\omega) \subseteq E$.

In words: whether or not C is cb, he can always ~~see~~ distinguish B, so he always knows B when it's true.

d) Let $E' = K_C(E)$. Then $K_A(E') = \{B-cb, B-ncb\}$.

A can tell when the source is B. And they know that in those worlds, C will also be able to determine that it's B (regardless of cb). So whenever B obtains, A knows C knows B.

Moreover, since by positive introspection $K_A(E')$, E' is self-evident: anytime it occurs, both players know it does.

And since $E' \subseteq E$, it's common knowledge in B-worlds that C knows it's a B-world.

e) Let $\tilde{E} = \{Ch-cb, Ch-ncb\}$. $K_C(\tilde{E}) = \{Ch-ncb\}$ i.e. C knows Ch only when it obtains and he's not colour blind.

f) Let $\tilde{E}' = K_C(\tilde{E})$. $K_A(\tilde{E}') = \emptyset$ since $\nexists \omega \in \Omega$ s.t. $P_A(\omega) \subseteq \tilde{E}'$.

A doesn't know whether C's acb. So she can't know ~~whether~~ he knows Ch (or even if Ch holds, which is required for C to know it).